Trehalose Preserves Soluble LmOx During Repeated Freeze-Thaw Cycles Without Increasing Catalytic Turnover

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ABSTRACT

The fictional oxidase LmOx loses soluble protein during repeated freeze-thaw handling. We compared 0%, 5%, 10%, and 20% trehalose over five cycles. Ten percent trehalose preserved 84% of the initial soluble LmOx, compared with 41% without trehalose. However, specific activity of the remaining soluble enzyme was unchanged at 118-123 U/mg across all groups. A pilot worksheet listed sucrose, pH 6.0, and 30 C incubation as candidate conditions, but these were not used in the reported experiment. Trehalose therefore improved physical recovery rather than catalytic turnover.

Keywords: *trehalose, freeze-thaw, protein stability, specific activity, negative result*

INTRODUCTION

Loss of soluble protein during freezing can be mistaken for loss of intrinsic catalytic quality. Measuring both total recovered activity and specific activity separates these outcomes.

This study tested trehalose concentration during five freeze-thaw cycles. Candidate sucrose and temperature conditions appeared in preliminary planning notes but were intentionally excluded from the final design. The reported experiment used pH 7.5 and thawing at 20 C for every group.

MATERIALS AND METHODS

Final experimental design

Purified LmOx at 0.5 mg/mL was prepared in 25 mM HEPES, pH 7.5, 150 mM NaCl with 0%, 5%, 10%, or 20% trehalose. Samples were frozen at -80 C for 2 h and thawed at 20 C for 30 min. Five cycles were completed.

Excluded pilot candidates

A pre-experiment worksheet contained possible tests at pH 6.0, 30 C, and with 10% sucrose. These conditions were not included in sample preparation, randomization, or data analysis. They are stated here only to preserve the audit trail.

Measurements

Soluble recovery was measured after centrifugation by absorbance and confirmed by SDS-PAGE. Catalytic activity was measured with a chromogenic substrate. Specific activity was calculated from the soluble protein remaining after each cycle.

Group	Trehalose	pH	Thaw temperature
T0	0%	7.5	20 C
T5	5%	7.5	20 C
T10	10%	7.5	20 C
T20	20%	7.5	20 C

Table 1. Final study groups. Sucrose, pH 6.0, and 30 C were not tested.

RESULTS

Soluble recovery

After five cycles, soluble LmOx recovery was 41 +/- 6% without trehalose, 68 +/- 5% with 5%, 84 +/- 4% with 10%, and 82 +/- 7% with 20% trehalose. Ten percent was selected as the lowest concentration reaching the recovery plateau.

Catalytic turnover did not increase

Specific activity after the fifth cycle was 118 +/- 8, 121 +/- 7, 123 +/- 6, and 119 +/- 9 U/mg for 0%, 5%, 10%, and 20% trehalose, respectively. These values were not meaningfully different. Total activity per original sample volume increased because more soluble enzyme remained, not because each milligram was more active.

Visual inspection

Visible precipitate was common in the 0% group and uncommon in the 10% group. No color change or substrate-independent absorbance increase was observed.

Trehalose	Soluble recovery	Specific activity (U/mg)
0%	41 +/- 6%	118 +/- 8
5%	68 +/- 5%	121 +/- 7
10%	84 +/- 4%	123 +/- 6
20%	82 +/- 7%	119 +/- 9

Table 2. Trehalose preserved soluble recovery but did not increase specific activity.

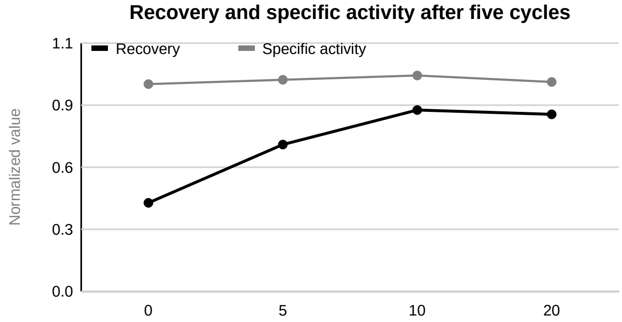

Figure 1. Recovery improved to a plateau while catalytic quality remained stable.

DISCUSSION

Trehalose reduced freeze-thaw precipitation and preserved soluble LmOx. The stable specific activity shows that trehalose did not activate the enzyme. The increase in total recovered activity resulted from retaining more soluble protein.

The excluded pilot conditions are a deliberate source of possible misreading. They should not be reported as experimental groups or findings.

LIMITATIONS

Only five cycles and one protein concentration were examined. Long-term storage, sucrose, pH 6.0, and thawing at 30 C were not tested.

CONCLUSION

Ten percent trehalose preserved 84% of soluble LmOx after five freeze-thaw cycles, compared with 41% without trehalose, while specific activity remained unchanged.

REFERENCES

The following references are fictional and are included only to test reference-section handling.

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